

Installation Data Sheet
Series: Direct Drive HSD 60Hz
Document No.: TI-DATA-2016-HSD 500 550 600 650
Version: 2.7
Revision Date: 07/29/2024

Model	HSD 500			HSD 550			HSD 600			HSD 650		
Rated Pressure [psig]	100	125	175	125	175	217	125	175	217	125	175	217

I. COOLING DATA

Cooling System Available [Std., Opt.]	W/C	W/C	W/C	W/C
Standard Ambient Temp. Range [°F]	40 - 115	40 - 115	40 - 115	40 - 115

VENTILATION OF COMPRESSOR ROOM

Air Inlet Opening [sq. ft. free area] (W/C) Z	14	15	16.2	18.3
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Solution A (forced ventilation with exhaust fan) as shown in service manual

Cooling Fan Capacity [CFM] (W/C)	10,594	11,772	12,949	14,126
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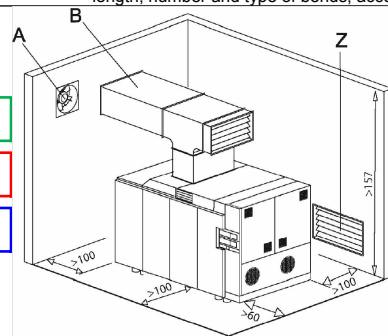
Solution B (exhaust air used for space heating) as shown in service manual

Internal Cooling Fan Capacity [CFM] (W/C)	5,886	5,886	5,886	5,886
Max. Additional Pressure Drop for Ducts [inch Water Column] (W/C)	0.16	0.16	0.16	0.16

Exhaust Air Opening Reference Dimensions (L x W) [in]	36 x 67	36 x 67	36 x 67	36 x 67
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See drawing for actual dimensions. The actual individual duct dimension will vary for every installation based on actual length, number and type of bends, accessories etc.

Model shown for reference only
Actual Duct size may vary with installation



Solution A Exhaust Fan

Solution B Exhaust Duct

Ventilation of Compressor Room Z

WATER COOLED DATA

Type of heat exchangers	stainless steel, plate-type	stainless steel, plate-type	stainless steel, plate-type	stainless steel, plate-type
Internal Cooling Fan Capacity [CFM]	5,886	5,886	5,886	5,886
Approach Temp. [°F]	1.8	1.8	1.8	1.8
Typical Heat Rejected into Cooling Water [BTU / HR]	1,282,000	1,419,500	1,593,500	1,767,500
Heat Rejected into Cooling Air [BTU / HR]	72,000	80,000	90,000	94,000
Max. outlet temperature [°F]	130	130	130	130

Reference conditions: 14.5 psia, 30% relative humidity and 68°F inlet air temperature.

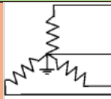
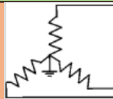
Based on highest input kW of machine.

Discharge temperature limited for non-treated water (to prevent calcification).

Examples for 2 temperature rises between water inlet and water discharge temperature [°F]	27	54	27	54	27	54	27	54
Max. inlet water temperature [°F]	105	68	105	68	105	68	105	68
Min. cooling water flow for Aggregate A/B [gpm]	53 / 44	26.5 / 22	53 / 53	26.5 / 26.5	66 / 53	33 / 26.5	66 / 66	33 / 33
Total Min. cooling water flow for Aggregate A/B [gpm]	97	48.5	106	53	119	59.5	132	66
Pressure drop across compressor package [psi] WITHOUT cooling water throttling valve	10 / 7.5	3.0 / 3.0	10 / 10	3.0 / 3.0	14.5 / 10	3.0 / 3.0	14.5 / 14.5	3.0 / 3.0
Pressure drop across compressor package [psi] WITH cooling water throttling valve	14 / 10	4.0 / 4.0	14 / 14	4.0 / 4.0	20 / 14	4.0 / 4.0	20 / 20	4.0 / 4.0



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II. ELECTRICAL DATA														
Electrical data may vary in accordance with motor manufacturer's specifications. Motors are EISA compliant.														
DRIVE MOTOR														
Motor HP		500			550			650			700			
Insulation Class		Motor A F			F			F			F			
Standard Voltage		Motor A 460V/3ph/60Hz			460V/3ph/60Hz			460V/3ph/60Hz			460V/3ph/60Hz			
Full Load Amps [FLA] @ 460V/3ph/60Hz		Motor A 305			305			380			380			
Full Load Amps [FLA] @ 575V/3ph/60Hz		Motor A 245			245			305			305			
Insulation Class		Motor B F			F			F			F			
Standard Voltage		Motor B 460V/3ph/60Hz			460V/3ph/60Hz			460V/3ph/60Hz			460V/3ph/60Hz			
Full Load Amps [FLA] @ 460V/3ph/60Hz		Motor B 305			305			305			390			
Full Load Amps [FLA] @ 575V/3ph/60Hz		Motor B 245			245			245			310			
FAN MOTOR (W/C)														
Insulation Class		F			F			F			F			
Fan Motor [HP], Single Speed @ 460V/3ph/60Hz		2 x 0.4			2 x 0.4			2 x 0.4			2 x 0.4			
Fan Motor [HP], Single Speed @ 575V/3ph/60Hz		2 x 0.8			2 x 0.8			2 x 0.8			2 x 0.8			
Full Load Amps [FLA] @ 460V/3ph/60Hz		1.2			1.2			1.2			1.2			
Full Load Amps [FLA] @ 575V/3ph/60Hz		2.4			2.4			2.4			2.4			
TOTAL PACKAGE DATA (W/C)														
Do NOT operate package on any unsymmetrical power supply. Also do NOT operate package on power supplies like, for example, a three-phase (open) delta or three-phase star with non-grounded neutral. The machine requires a symmetrical three-phase power supply transformer with a WYE configuration output as shown on the right. In a symmetrical three-phase supply the phase angles and voltages are all the same. Other power supplies are not suitable.							three-phase star (wye); 4-wire; grounded neutral						three-phase star (wye); 3-wire; grounded neutral	
Continuous Duty [Hours per day]				24			24			24			24	
Control Cabinet Class (NEMA)				12			12			12			12	
Short Circuit Current Rating (SCCR) [kA] @ 460V/3ph/60Hz		Field installed fuse required, see below*		50			50			50			50	
Short Circuit Current Rating (SCCR) [kA] @ 575V/3ph/60Hz		Field installed fuse required, see below*		50			50			50			50	
Package Full Load Amps @ 460V/3ph/60Hz [FLA]				587			648			727			807	
Package Full Load Amps @ 575V/3ph/60Hz [FLA]				492			554			620			686	
Recommended Disconnect Fuse Size [Amps] @ 460V/3ph/60Hz		*Time delay (dual element) fuse; Class J ≤ 600A (e.g. AJT) / Class L > 600A (e.g. A4BQ). Based on 2020 NEC 240.6, 430.52, and Tables 430.52, 430.248, and 430.250		800			800			1000			1200	
Recommended Disconnect Fuse Size [Amps] @ 575V/3ph/60Hz				700			800			800			1000	
Recommended Disconnect Wire Size [AWG/kcmil] @ 460V/3ph/60Hz		The following multi-strand copper core wires are given according to 2020 NEC 310.14, 310.15, 310.16 and table 310.16 adjusted for 40°C ambient temperature. If other local conditions prevail, for example high temperature, the cross section should be checked and adjusted according to 2020 NEC 110.14(C), 220.3, 310.14, 310.15, 310.16, 430.6, 430.22, 430.24, 670.4(A) and other local codes.		4 x 4/0 AWG per phase and ground			4 x 250 kcmil per phase and ground			4 x 300 kcmil per phase and ground			4 x 350 kcmil per phase and ground	
Recommended Disconnect Wire Size [AWG/kcmil] @ 575V/3ph/60Hz				3 x 250 kcmil per phase and ground			3 x 300 kcmil per phase and ground			3 x 350 kcmil per phase and ground			3 x 400 kcmil per phase and ground	



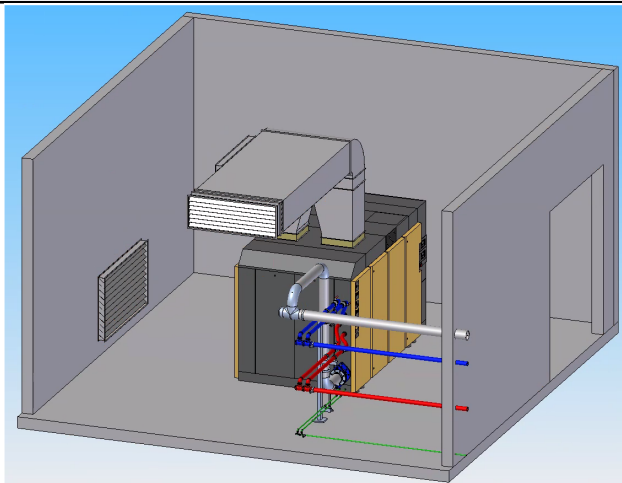
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INSTALLATION and MAINTENANCE DATA													
W/C with Super Soundproofing [dB(A)]		SOUND PRESSURE LEVEL [Measured in dB(A) according to ISO 2151 using ISO 9614-2]			73		74		74		75		
W/C Air Discharge [inches NPT or Flange]		6 ANSI/ASME B16.5 class 150											
Cooling Water Connection [inches NPT or Flange]		4 x 2 NPT											
Power Input Conduit Opening(s) [inches]		3 (x 4)			3 (x 4)			3 (x 4)			3 (x 4)		
Condensate Drain Connection [NPT]		2 @ 1/2			2 @ 1/2			2 @ 1/2			2 @ 1/2		
Width [inches]		140 1/2			140 1/2			140 1/2			140 1/2		
Depth [inches]		84 1/2			84 1/2			84 1/2			84 1/2		
Height [inches]		92 1/2			92 1/2			92 1/2			92 1/2		
Floor Space [sq. ft.]		82 4/9			82 4/9			82 4/9			82 4/9		
Weight (W/C) [lb]		Weight may vary based on airend selected.			16,493		18,082		18,594		19,105		
COMPRESSOR FLUID DATA													
Fluid Capacity (W/C) [gal]		2 x 31.7			2 x 31.7			2 x 31.7			2 x 31.7		
Flow Rate [gal/min]		2 x 73			2 x 73			2 x 73			2 x 73		
Typical Oil Consumption [fl. Oz./100 h]		57.6			62.4			68.4			74.3		
Standard Fluid Type		Sigma S-460			Sigma S-460			Sigma S-460			Sigma S-460		
MAINTENANCE PARTS													
Air Inlet Filter		4E0304.0 (x2)											
Filter Mat for Control Cabinet		7.4519.0 x 4											
Fluid Filter		6.4693.0 x 6											
Fluid Separator Kit		6.3559.00010 x 2											

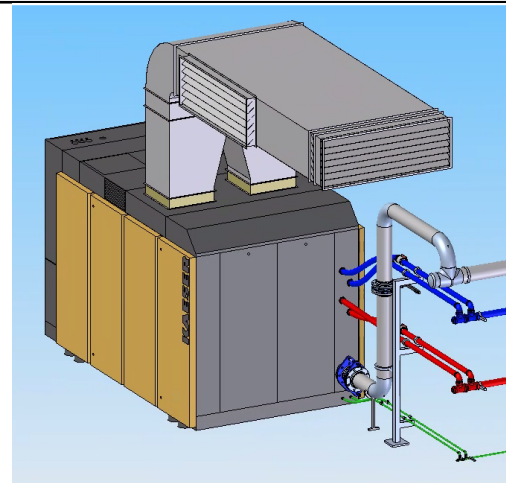
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SAMPLE SKETCHES

Sample Installation Planning
Examples of room ventilation and ductwork
Please note the upsizing required for compressor exhaust ducts



HSD fixed speed



Example designs only, not for construction purposes.